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Chapter 8 — Plant Physiology

Complete Chapter Notes · Textbook topics converted into board-style long questions, with important conceptual short questions inside every topic · PECTAA 2025–26, Punjab English Medium

This chapter follows the textbook topics 8.1 to 8.12. Each topic is converted into board-style long questions, and each topic ends with conceptual short questions. Direct definitions are omitted.

Board pattern (Punjab 2026, new syllabus): Chapter 8 carries 2 MCQs, 4 short questions in Q.2 (the highest short-question weight), and part (b) of a long question in Q.9. Shorts are 2 marks; longs are 8 marks.

TOPIC 8.1 — NUTRITION IN PLANTS

Textbook topic 8.1 · printed pages 177–180

Q.1 Differentiate between macronutrients and micronutrients. Describe the roles of important plant nutrients.

Board long question · 8 marks

All autotrophic organisms need carbon dioxide and water, which supply carbon, oxygen and hydrogen. Other nutrients are obtained from the environment and are divided into two groups.

Group	Required amount	Elements
Macronutrients	Relatively larger amounts	Carbon, hydrogen, oxygen, nitrogen, potassium, calcium, phosphorus, magnesium, sulphur
Micronutrients	Very small amounts	Iron, manganese, zinc, molybdenum, copper, chlorine, boron

- **Nitrogen:** Essential for energy metabolism and protein production; deficiency causes leaf loss and stunted growth.
- **Phosphorus:** Part of ATP; promotes root growth and flowering; deficiency delays flowering and causes browning/wrinkling of leaves.
- **Potassium:** Involved in water regulation and transport of reserve substances; deficiency causes dark patches on leaves.
- **Calcium:** Provides stability to cell walls; deficiency causes yellow/brown patches.
- **Magnesium:** Core of chlorophyll; deficiency causes loss of greenness in older leaves.
- **Sulphur:** Important in nitrogen metabolism; deficiency makes plant lighter in colour.
- **Iron:** Essential for chlorophyll synthesis and acts as enzyme cofactor.

- **Manganese:** Important for antioxidant enzymes that mitigate oxidative stress.

Q.2 How do insectivorous plants supplement their nutrition?

Board long question · 8 marks

Some plants supplement organic molecules into their food in addition to inorganic nutrients. These organic chemicals are acquired by capturing and breaking down insects and tiny animals. All insectivorous plants are true autotrophs, but their development accelerates when they capture prey, apparently because nitrogenous compounds of animal body are beneficial.

- **Pitcher plant:** Leaves are modified into a sac or pitcher partly filled with water. The terminal portion forms a hood, and stiff hairs prevent insects from crawling out.
- **Venus fly trap:** Has two lobes hinged at the end of each leaf. Trichomes on the inner surface trigger rapid closure when prey contacts them. Fine bristles interlock to prevent escape, and enzymes digest the trapped insect.
- **Sundew:** Leaves are covered with tiny hairs resembling tentacles, each with a drop of sticky "dew". Insects attracted by odour are trapped, digested by enzymes and absorbed.

Important Conceptual Short Questions — Topic 8.1

S1. Why do plants require both macronutrients and micronutrients?

Macronutrients are needed in larger amounts for structure and metabolism, while micronutrients are needed in small amounts as enzyme cofactors and for specific physiological roles.

S2. How does magnesium deficiency affect photosynthesis?

Magnesium is the core of the chlorophyll molecule. Its deficiency reduces chlorophyll, causing loss of greenness and reduced photosynthesis.

S3. Why is nitrogen essential for plant growth?

It plays an essential role in energy metabolism and protein production. Deficiency causes leaf loss and stunted growth.

S4. Why are insectivorous plants still considered autotrophs?

They photosynthesize and make their own food. Insect capture only supplements nitrogenous compounds and accelerates development.

S5. How does the Venus fly trap ensure that prey cannot escape?

Trichomes trigger rapid lobe closure, and fine bristles interlock upon closure, trapping the insect inside for digestion.

TOPIC 8.2 — GAS EXCHANGE IN PLANTS

Textbook topic 8.2 · printed pages 180–182

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Q.3 Explain the opening and closing of stomata. Describe the starch-sugar hypothesis and the K^+ ion influx mechanism.

Board long question · 8 marks

Stomata are tiny openings or pores necessary for gaseous exchange, typically found in leaves and sometimes stems. They are surrounded by **guard cells**, which are bean-shaped and contain chloroplasts. Guard cells open and close the stomatal pore depending on environmental conditions, controlling transpiration rate.

Starch-sugar hypothesis

Proposed by H. Van Mohl in 1856. During the day, guard cells photosynthesize and produce sugars, lowering their water potential. Water enters, guard cells become turgid and stomata open. At night, sugars are converted to starch or used in respiration, water moves out, guard cells become flaccid and stomata close. This mechanism does not fully explain rapid turgor changes.

Influx of K^+ ions

Active transport of K^+ into guard cells reduces their osmotic potential. Water enters by osmosis, guard cells become turgid and stomata open. Blue light enhances K^+ uptake. At night, K^+ diffuses out, water is lost and stomata close. At high temperature, when leaf cells wilt, **abscisic acid** is released and stops K^+ pumping, causing stomatal closure.

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Important Conceptual Short Questions — Topic 8.2

S1. How do guard cells control the rate of transpiration?

By opening and closing the stomatal pore, they regulate water vapour loss and gas exchange.

S2. Why does sugar accumulation in guard cells cause stomatal opening?

It lowers water potential, so water enters by osmosis, making guard cells turgid and opening the pore.

S3. How does K^+ influx lead to stomatal opening?

Active uptake of K^+ lowers osmotic potential, water enters by osmosis and guard cells become turgid.

S4. Why does abscisic acid cause stomatal closure during wilting?

It stops the active transport of K^+ into guard cells, so water leaves and guard cells become flaccid, closing the stomata.

S5. Why do stomata generally close at night?

Photosynthesis ceases, sugars are converted to starch or used in respiration, and K^+ diffuses out, causing water loss and flaccidity of guard cells.

TOPIC 8.3 — SUPPORT IN PLANTS

Textbook topic 8.3 · printed pages 182–183

Q.4 Describe the supporting tissues in plants.

Board long question · 8 marks

Tissue	Structure	Role in support
Parenchyma	Living cells; absorb water and create turgor pressure	Supports herbaceous plants and soft parts through turgidity
Collenchyma	Living cells with primary walls thickened at corners; no secondary walls	Supports young stems, leaf stalks and veins; elongates with growth
Sclerenchyma	Usually dead cells with thick lignified secondary walls	Provides rigid support; includes fibres, sclereids and vessels
Xylem	Lignified tracheids and vessels	Conducts water and provides mechanical support
Phloem	Sieve tubes and companion cells	Transports food; fibres provide some support

Turgor pressure arises from osmotic pressure within the vacuole. The **tonoplast** actively moves ions into the vacuole, drawing water in and making cells turgid, which supports soft tissues.

Important Conceptual Short Questions — Topic 8.3

S1. How does turgor pressure provide support to herbaceous plants?

Water entering the vacuole creates internal hydrostatic pressure against the cell wall, making cells rigid and supporting soft tissues.

S2. Why is collenchyma suitable for supporting young growing parts?

Its cells have flexible primary walls thickened at corners and can elongate as the stem or leaf grows, providing support without restricting growth.

S3. How does lignin strengthen sclerenchyma?

Lignin saturates the thick secondary walls, making them rigid and strong, suitable for mechanical support.

S4. Differentiate between fibres and sclereids.

Fibres are elongated cylindrical cells found in bundles, while sclereids are smaller and found in seed coats and nut shells, offering protection.

TOPIC 8.4 — WATER POTENTIAL

Textbook topic 8.4 · printed pages 183–184

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Q.5 Explain water potential and its components. How does it determine the direction of water movement?

Board long question · 8 marks

Water molecules possess kinetic energy. The greater the concentration of water molecules in a system, the greater the total kinetic energy, called **water potential** (Ψw). Pure water has maximum water potential, defined as zero. Water moves from higher Ψw to lower Ψw.

In plant cells, water potential is determined by two components: **solute potential (Ψs)** and **pressure potential (Ψp)**. The relationship is: **Ψw = s + Ψp**.

- **Solute potential (Ψs):** Also called osmotic potential. It is always negative. More solute molecules make Ψs more negative, lowering water potential.
- **Pressure potential (Ψp):** Due to pressure exerted by water. When water enters plant cells by osmosis, pressure builds up, making the cell turgid and increasing Ψp.

Osmosis is the movement of water from higher water potential to lower water potential through a partially permeable membrane. The steeper the potential gradient, the faster the flow of water.

Important Conceptual Short Questions — Topic 8.4

S1. Why does adding solute lower water potential?

Solute molecules reduce the free energy and concentration of water molecules, making solute potential more negative.

S2. How does pressure potential increase when a plant cell becomes turgid?

Water entering by osmosis builds up pressure against the cell wall, increasing pressure potential.

S3. Why does water move from soil to root and from root to leaf?

It moves along a water potential gradient, from higher water potential in soil to lower water potential in root and leaf cells.

S4. How is osmosis related to water potential?

Osmosis is the movement of water from higher water potential to lower water potential through a partially permeable membrane.

TOPIC 8.5 — TRANSPORT OF WATER IN PLANTS

Textbook topic 8.5 · printed pages 184–187

Q.6 Describe the uptake and transport of water in plants. Explain the TACT mechanism.

Board long question · 8 marks

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Uptake by roots

Root hairs, extensions of epidermal cells, provide a large surface area for absorption. Water and minerals enter root epidermal cells by active and passive transport and move to the xylem through three pathways:

- **Apoplast pathway:** Through adjacent cell walls; discontinuous at the endodermis due to Casparian strips.
- **Symplast pathway:** Through interconnected protoplasts via plasmodesmata.
- **Vacuolar pathway:** Through cell membranes, cytoplasm, tonoplast and vacuoles; negligible.

Xylem structure

Xylem consists of tracheids, vessels, xylem fibres and xylem parenchyma. Tracheids are elongated cells with bordered pits; vessels are broader with perforation plates; fibres provide support; parenchyma stores nutrients and helps lateral transport.

TACT mechanism

- **Transpiration:** Evaporation from leaves creates negative pressure, pulling water upward.
- **Adhesion:** Water molecules stick to xylem walls, maintaining the water column.
- **Cohesion:** Hydrogen bonds keep water molecules together in an unbroken column.
- **Tension:** The pulling force from transpiration extends through the xylem, drawing water against gravity.

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Important Conceptual Short Questions — Topic 8.5

S1. Why are root hairs important for water absorption?

They greatly increase the surface area of the root epidermis, allowing more water and mineral uptake.

S2. How does the Casparian strip affect the apoplast pathway?

It blocks movement through cell walls in the endodermis, forcing water and minerals to cross cell membranes and enter the symplast.

S3. Differentiate between tracheids and vessels.

Tracheids are elongated, tapered cells with bordered pits, while vessels are shorter, broader and joined end-to-end with perforation plates for efficient water flow.

S4. How does transpiration pull water upward?

Evaporation from leaves creates negative pressure (tension) that pulls the cohesive water column up through the xylem.

S5. Why are cohesion and adhesion necessary for water transport?

Cohesion keeps water molecules together in a continuous column, while adhesion helps water stick to xylem walls, preventing breakage under tension.

TOPIC 8.6 — TRANSLOCATION OF FOOD IN PLANTS

Textbook topic 8.6 · printed pages 187–189

Q.7 Describe the structure of phloem and explain the pressure-flow theory of translocation.

Board long question · 8 marks

Structure of phloem

Phloem transports organic nutrients, particularly photosynthetic products, from leaves to other parts. It consists of **sieve tube elements**, **companion cells**, phloem fibres and phloem parenchyma. Sieve tube elements have sieve areas and form sieve tubes. Companion cells supply ATP and proteins to sieve tube elements through plasmodesmata.

Pressure-flow theory (Munch, 1930)

1. Glucose formed in mesophyll cells is converted to sucrose for transport.
2. Sucrose is actively loaded into companion cells and then sieve tubes, increasing solute concentration at the source.
3. Water enters sieve tubes from nearby xylem by osmosis, increasing pressure at the source end.
4. At the sink, sugar is actively unloaded and water follows by osmosis, lowering pressure.
5. The pressure difference drives mass flow of sucrose solution from source to sink.

Textbook wording says loading increases water potential at the source; the correct interpretation is that solute lowers water potential while pressure potential rises, driving flow from source to sink.

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Important Conceptual Short Questions — Topic 8.6

S1. Why is sucrose loaded into phloem at the source?

Loading increases solute concentration, causing water to enter by osmosis and build pressure that drives translocation.

S2. How do companion cells support sieve tube elements?

They supply ATP and proteins through plasmodesmata, helping sieve tube elements carry out active loading and transport.

S3. Why is diffusion insufficient to explain phloem transport?

Diffusion is far too slow (about 1 m per 8 years) compared with the observed rate of sugar movement (about 1 m per hour).

S4. How does the pressure-flow theory explain movement from source to sink?

Sugar loading at the source raises pressure, while unloading at the sink lowers pressure, so the solution flows along the pressure gradient.

S5. Why does water follow sugar during unloading at the sink?

Removal of sugar lowers solute concentration, so water moves out by osmosis, reducing pressure at the sink.

TOPIC 8.7 — GROWTH IN PLANTS

Textbook topic 8.7 · printed pages 189–194

Q.8 Describe meristems and the types of growth in plants. How are annual rings formed?

Board long question · 8 marks

Types of meristems

Meristem	Location	Function
Apical	Tips of roots and shoots	Primary growth; increase in length
Intercalary	Bases of internodes, e.g., grasses	Temporary growth; production of leaves and flowers
Lateral	Vascular cambium and cork cambium	Secondary growth; increase in thickness

Primary growth

Primary growth increases length and occurs in three phases: **cell division** in the apical meristem, **cell elongation** by water uptake, and **cell differentiation** into specialized tissues.

Secondary growth

Secondary growth increases thickness and is due to lateral meristems. Vascular cambium produces secondary xylem inward and secondary phloem outward. Cork cambium produces cork, forming bark.

Annual rings

In spring, active cambium produces wide, light-coloured **spring wood**. In autumn, reduced activity produces narrow, dense **autumn wood**. Together they form an annual ring, used in **dendrochronology** to determine tree age and past climate.

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Q.9 Describe the major plant growth regulators and their functions.

Board long question · 8 marks

Regulator	Major functions
Auxins	Cell enlargement, apical dominance, root initiation, fruit growth, parthenocarpy, inhibit abscission
Cytokinins	Cell division, delay senescence, promote bud initiation
Gibberellins	Stem elongation, seed germination, flowering, fruit growth
Abscisic acid	Stomatal closure, dormancy, stress response
Ethylene	Fruit ripening, senescence, abscission

Important Conceptual Short Questions — Topic 8.7

S1. How do apical and lateral meristems differ in function?

Apical meristems increase length through primary growth, while lateral meristems increase thickness through secondary growth.

S2. Why are intercalary meristems important in grasses?

They are located at the bases of internodes and allow regrowth of leaves and stems after grazing or cutting.

S3. How does cell elongation contribute to primary growth?

Newly formed cells take up water, increase in volume and attain final size, extending the plant body.

S4. Why do annual rings form in trees?

Seasonal cambial activity produces wide spring wood and dense autumn wood each year, creating visible rings.

S5. How can dendrochronology reveal past climate conditions?

The width and density of annual rings reflect growing conditions each year, allowing reconstruction of past climate.

S6. How do auxins promote apical dominance?

They stimulate growth of the apical bud while inhibiting lateral buds, so the main shoot grows more strongly.

S7. Why is abscisic acid called a stress hormone?

It promotes stomatal closure and dormancy, helping plants survive water stress and unfavourable conditions.

S8. How does ethylene affect fruit ripening?

It triggers biochemical changes that soften fruit, convert starches to sugars and develop colour and aroma.

TOPIC 8.8 — OSMOREGULATION IN PLANTS

Textbook topic 8.8 · printed pages 194–197

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Q.10 Explain osmoregulation in plants. How do hydrophytes, mesophytes, xerophytes and halophytes adapt to their habitats?

Board long question · 8 marks

Osmoregulation is the maintenance of a stable internal equilibrium of water and dissolved substances. Solutions may be **hypotonic** (water enters cell), **hypertonic** (water leaves cell, causing plasmolysis) or **isotonic** (no net movement).

Plant type	Habitat	Adaptations
Hydrophytes	Aquatic	Thin/absent cuticle, reduced roots, hydathodes or vacuoles to expel excess water; marine forms excrete salts and synthesize organic solutes
Mesophytes	Moderate	Well-developed roots/shoots, broad leaves, stomata; no special adaptations
Xerophytes	Dry	Waxy cuticle, leaf dropping/folding, succulent water-storing stems
Halophytes	Saline	Succulent leaves/stems, salt tolerance, sometimes leaves modified into spines; mangroves near seashores

Important Conceptual Short Questions — Topic 8.8

S1. Why does a plant cell become plasmolysed in a hypertonic solution?

Water leaves the cell by osmosis, causing the protoplast to shrink away from the cell wall.

S2. How do xerophytes reduce water loss?

They have waxy cuticles, reduced or folded leaves, leaf dropping and water-storing succulent tissues.

S3. Why do halophytes need special adaptations?

They live in saline soils where high salt concentration makes water uptake difficult, so they must tolerate or excrete salts and maintain osmotic balance.

S4. How do freshwater hydrophytes avoid overhydration?

They expel excess water through hydathodes or vacuoles and actively absorb ions to maintain osmotic balance.

S5. Differentiate between hydrophytes and xerophytes.

Hydrophytes live in aquatic environments and have adaptations for excess water, while xerophytes live in dry environments and have adaptations to conserve water.

TOPIC 8.9 — THERMOREGULATION IN PLANTS

Textbook topic 8.9 · printed pages 197–198

Q.11 How do plants cope with high and low temperature stress?

Board long question · 8 marks

High temperature

High temperature denatures enzymes and damages metabolism. Plants use **evaporative cooling** through transpiration. In heat stress, cells synthesize **heat-shock proteins** that embrace enzymes and other proteins, preventing denaturation.

Low temperature

Low temperature alters membrane fluidity as lipids crystallize. Plants increase the proportion of **unsaturated fatty acids** to prevent crystal formation and maintain membrane structure. Rapid chilling is more stressful than gradual cooling because this adaptation requires time.

Freezing can cause ice crystals. Ice formation in cell walls is less harmful, but ice within the protoplasm perforates membranes and organelles, killing cells. Cold-region plants adjust solute composition so the cytosol can **supercool** without ice formation.

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Important Conceptual Short Questions — Topic 8.9

S1. How do heat-shock proteins protect plants from high temperature?

They bind to enzymes and other proteins, preventing denaturation and maintaining metabolic function under heat stress.

S2. Why do plants increase unsaturated fatty acids in cold conditions?

Unsaturated fatty acids prevent membrane lipids from crystallizing, maintaining membrane fluidity and transport at low temperature.

S3. Why is rapid chilling more damaging than gradual cooling?

Plants need time to adjust membrane composition and solute concentration; rapid chilling does not allow these protective adaptations.

S4. Why is intracellular ice formation more dangerous than extracellular ice?

Ice crystals inside the protoplasm perforate membranes and organelles, killing the cell, while ice in cell walls is less damaging.

TOPIC 8.10 — MOVEMENTS IN PLANTS

Textbook topic 8.10 · printed pages 198–199

Q.12 Describe tropic movements in plants.

Board long question · 8 marks

Plants are fixed, so they respond to stimuli by altering growth patterns. **Tropic movements** are growth movements triggered by stimuli, occurring as curvature toward or away from the stimulus.

Tropism	Stimulus	Example
Phototropism	Light	Shoot tips show positive phototropism; roots show negative phototropism
Geotropism	Gravity	Roots show positive geotropism; shoots show negative geotropism
Thigmotropism	Touch	Climbing vines coil around supports
Chemotropism	Chemicals	Fungal hyphae grow toward chemical signals

Important Conceptual Short Questions — Topic 8.10

S1. How does phototropism help shoots obtain light?

Differential growth causes shoots to bend toward light, maximizing photosynthesis.

S2. Why do roots show positive geotropism?

They grow downward in response to gravity, anchoring the plant and accessing water and minerals in the soil.

S3. How does thigmotropism help climbing plants?

Contact with a support causes differential growth, allowing tendrils to coil around the support and climb.

S4. Differentiate between tropic movements and locomotion.

Tropic movements are growth curvatures in fixed plants, while locomotion is movement of the whole organism from place to place.

TOPIC 8.11 — PHOTOPERIODISM

Textbook topic 8.11 · printed pages 199–200

Q.13 Explain photoperiodism and the role of phytochrome in flowering.

Board long question · 8 marks

Photoperiodism is the response of plants to the length of day and night, enabling adaptation to seasonal changes. Garner and Allard (1920) showed that tobacco flowers only after exposure to short days. Plants are classified as **long-day plants**, **short-day plants** and **day-neutral plants**.

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Later studies showed that the **length of the dark period** is critical. Short-day plants are really long-night plants; interrupting the long night with light prevents flowering. Long-day plants flower if the long night is interrupted.

Phytochrome

Phytochrome exists in two forms: **P_{660}** (absorbs red light at 660 nm) and **P_{730}** (absorbs far-red light at 730 nm). Red light converts P_{660} to P_{730}; far-red light or darkness converts P_{730} back to P_{660}. The rate of conversion provides a "clock" for measuring darkness.

Red light inhibits flowering in short-day plants but promotes flowering in long-day plants, showing that phytochrome regulates the flowering response.

Important Conceptual Short Questions — Topic 8.11

S1. Why are short-day plants actually long-night plants?

They require a continuous long dark period to flower. Interrupting the night with light prevents flowering.

S2. How does phytochrome measure the length of darkness?

The conversion of P_{730} back to P_{660} in darkness occurs at a known rate, providing a biochemical clock for night length.

S3. Why does a flash of light during the night prevent flowering in short-day plants?

It converts P_{660} to P_{730}, interrupting the long dark period required for flowering.

S4. Differentiate between long-day and short-day plants.

Long-day plants flower when nights are short or interrupted, while short-day plants flower only when nights are long and uninterrupted.

S5. How does photoperiodism help plants reproduce at favourable times?

It synchronizes flowering with seasonal conditions, ensuring reproduction when pollinators and environmental conditions are suitable.

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TOPIC 8.12 — VERNALISATION

Textbook topic 8.12 · printed pages 200–201

Q.14 Explain vernalisation and its significance.

Board long question · 8 marks

Vernalisation is the stimulation of flowering in biennial and perennial plants by exposure to **low temperature**. The stimulus is received by the **shoot apex** of a mature stem or the embryo of the seed, not by leaves as in photoperiodism.

For some plants vernalisation is an absolute requirement; for others it assists flowering. The chilling treatment may last from four days to three months, with about **4°C** being very effective. It stimulates production of a hormone called **vernalin**, which induces flowering.

Photoperiodism and vernalisation synchronise reproductive behaviour with the environment, ensuring reproduction at favourable times and encouraging cross-pollination for genetic variability.

Important Conceptual Short Questions — Topic 8.12

S1. How does vernalisation differ from photoperiodism?

Vernalisation is induction of flowering by low temperature, perceived by the shoot apex or embryo. Photoperiodism is response to day/night length, perceived by leaves.

S2. Why is vernalisation important for biennial plants?

It ensures that flowering occurs only after exposure to winter cold, synchronizing reproduction with favourable seasonal conditions.

S3. How does vernalin induce flowering?

It is produced in response to low temperature and acts as a hormonal signal that stimulates flowering.

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